

Yena Kim

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Expected Graduation: Feb 2027

Education

Ewha Womans University

B.S. in Mathematics Education; Double Major in Statistics and Software; GPA: 4.08/4.5

Seoul, Republic of Korea

Mar 2022 - Feb 2027 (Expected)

San Diego State University

Exchange Student, Computer Science and Statistics; GPA: 3.88/4.00; 18 credits

San Diego, CA, USA

Aug 2025 - Dec 2025

Professional Experience

AI Research Intern

Brain Science Institute

Korea Institute of Science and Technology (KIST)

Jun 2026 - Aug 2026

- Evaluated the **cross-lingual robustness of 9 LLMs** across 5 psychological scales and 106 items, comparing response distributions, variability, entropy, and scale-level reliability between Korean and English prompts.
- Extended a biologically grounded neuron–astrocyte formulation from single-head to **multi-head attention**, derived its correspondence to standard MHA, and implemented the proposed architecture and comparative experiments in PyTorch.
- Reimplemented an **astromorphic Transformer** with collaborators and designed complexity-controlled lesion experiments to compare attention and FFN impairments across task complexity and training stages, observing distinct module-specific degradation patterns.

Research & Projects

Reliable Korean YouTube Misinformation Detection - Ewha Challenge Semester

Feb 2026 - Jun 2026

- Developed an **uncertainty-aware LLM framework** for Korean misinformation detection by combining repeated LLM reasoning with **conformal prediction**, enabling the model to abstain when confidence is insufficient.
- Designed a **TF-IDF reasoning-consistency score** from 7 repeated Qwen3.5-4B inferences on 1,435 transcripts, achieving **88.8% accuracy among selectively accepted predictions**; validated the method across randomized splits and deployed the pipeline as a React/FastAPI service.

Undergraduate Researcher & Team Lead - Ewha Data Science Lab

Jun 2025 - Aug 2025

- Led a team project using **2023 KNHANES data (n=4,358)**, merging health-examination and 24-hour dietary-recall data, engineering food-group intake ratios, and selecting features through ANOVA and VIF-based multicollinearity checks.
- Built multiclass diabetes-status models with Logistic Regression, Random Forest, and XGBoost; addressed class imbalance with SMOTE and selected XGBoost as the best model (**accuracy 0.74, macro ROC-AUC 0.891, macro F1 0.73**), then applied SHAP to interpret dietary and lifestyle risk factors.

Publications & Presentations

ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data

Kim, K., Kim, Y., & Seo, M. *International Conference on ICT Convergence (ICTC 2026)*, accepted; presentation scheduled Oct 2026.

- Designed a **context-aware LLM-GBDT fusion architecture** for sleep-state prediction, fine-tuning Qwen2.5-7B-Instruct on personalized behavioral context and integrating it with LightGBM, XGBoost, and CatBoost.
- Developed a **sample-wise reliability-guided fusion mechanism** that dynamically adjusts LLM contribution based on data quality and context availability.

Association Between Medical Factors and Employment Status in Workers' Compensation Insurance

2025

- Kim, Y. & Yoo, H. Poster presented at the Academic Conference on the Panel Study of Workers' Compensation Insurance.

Scholarships & Awards

Korea–U.S. Advanced Technology Youth Exchange Program Scholarship (2025) — MOTIE-funded nationwide STEM selection; approx. **USD 9,000**.

Ewha Challenge Semester Scholarships (2026) — **KRW 3.5M total**, including an Outstanding Project scholarship.

Research Assistant Scholarship, Ewha Womans University (2026) — **KRW 2.0M**; quantitative and textual data processing for International Studies research.

Murata Electronics Idea Contest 2025 - Award Recipient — proposed a text-to-physical-Braille accessibility device.

6th SW Convergence Major TELOS Track Competition - Honorable Mention (2025) — designed a gamified service for socially isolated youth.

Skills

Programming: Python, R, MATLAB **Machine Learning:** PyTorch, Hugging Face, scikit-learn, LightGBM, XGBoost, CatBoost

Data Analysis: pandas, NumPy, SHAP **Development/Tools:** FastAPI, React, Git/GitHub, VS Code

Spoken Languages: Korean (Native), English (Professional Working Proficiency; TOEFL iBT 100, TOEIC Speaking AL)

Certification: Advanced Data Analytics Semi-Professional (ADsP), Korea Data Agency